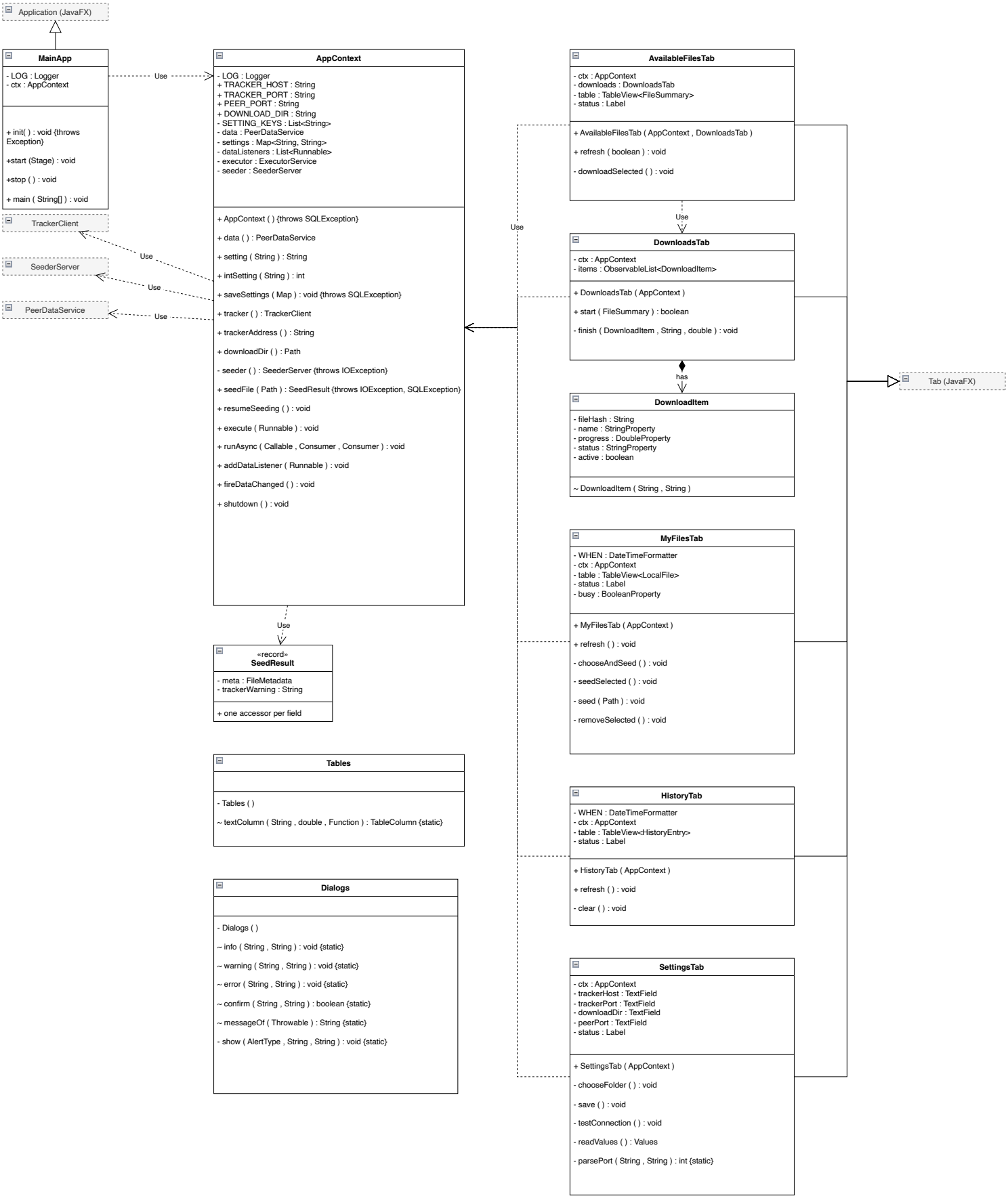
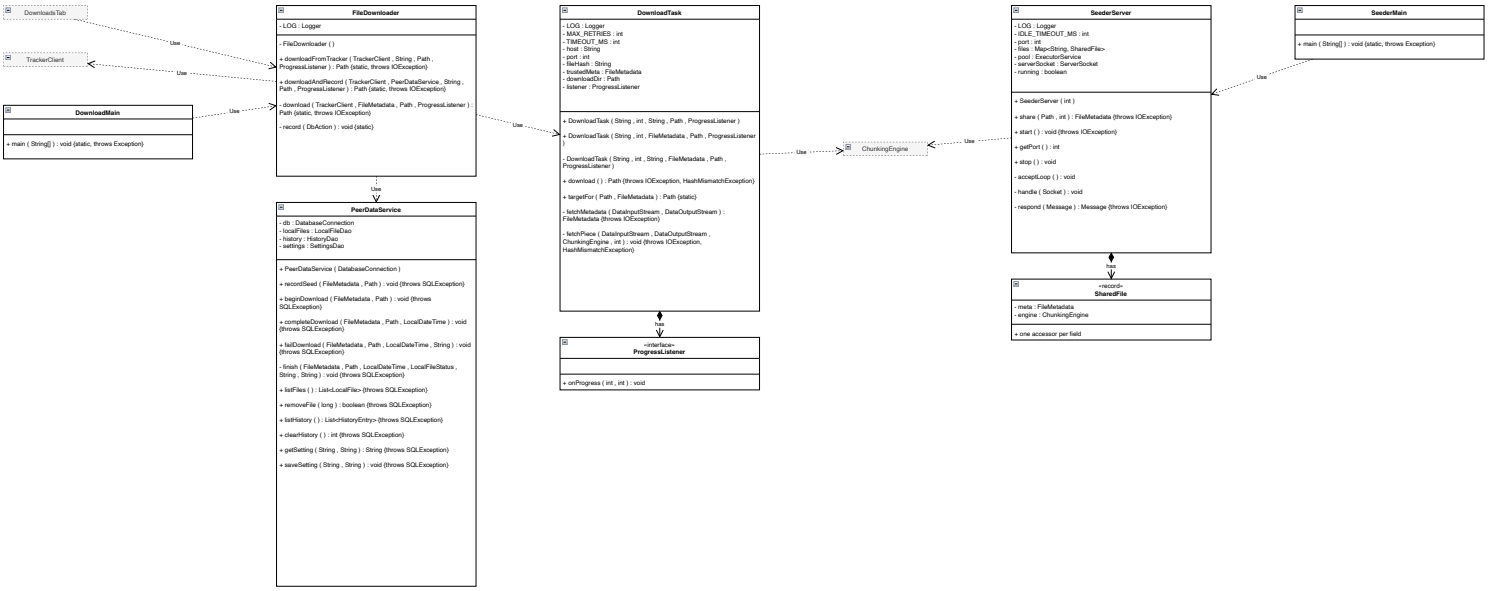
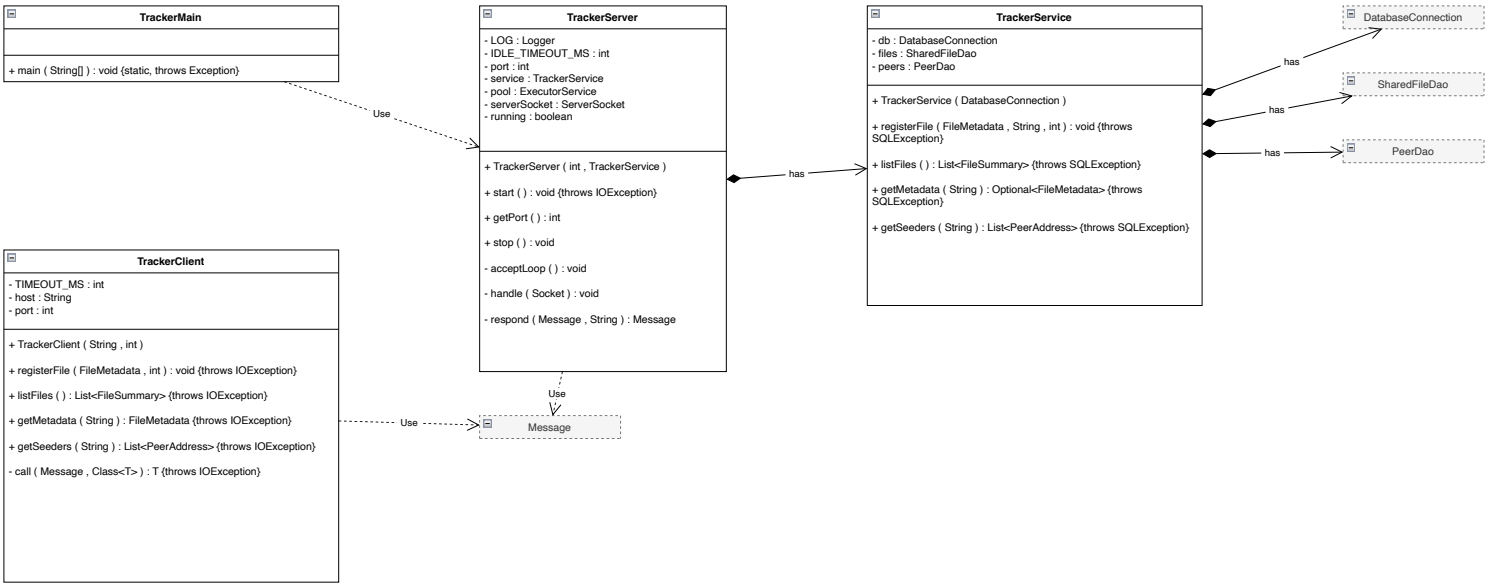
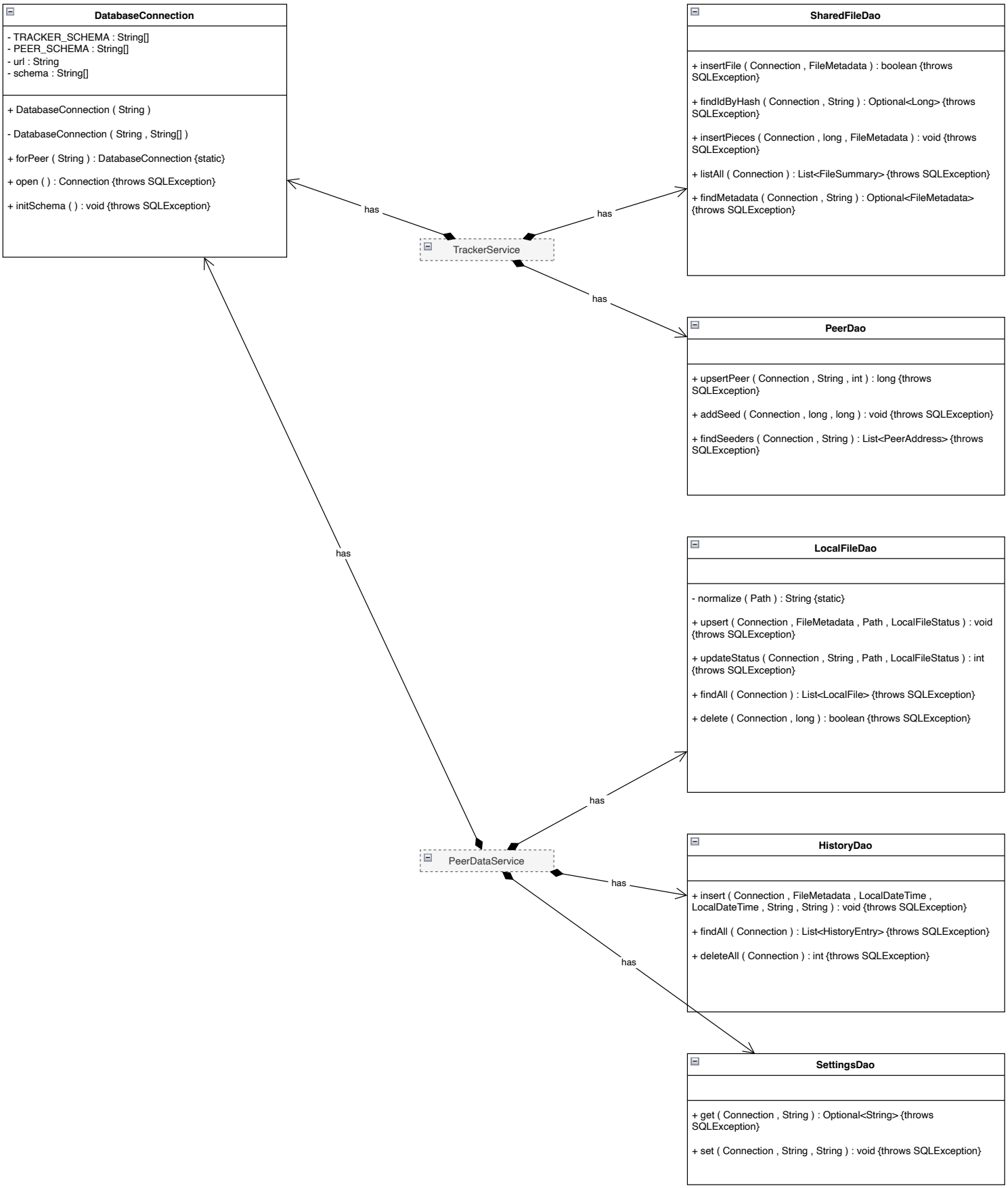
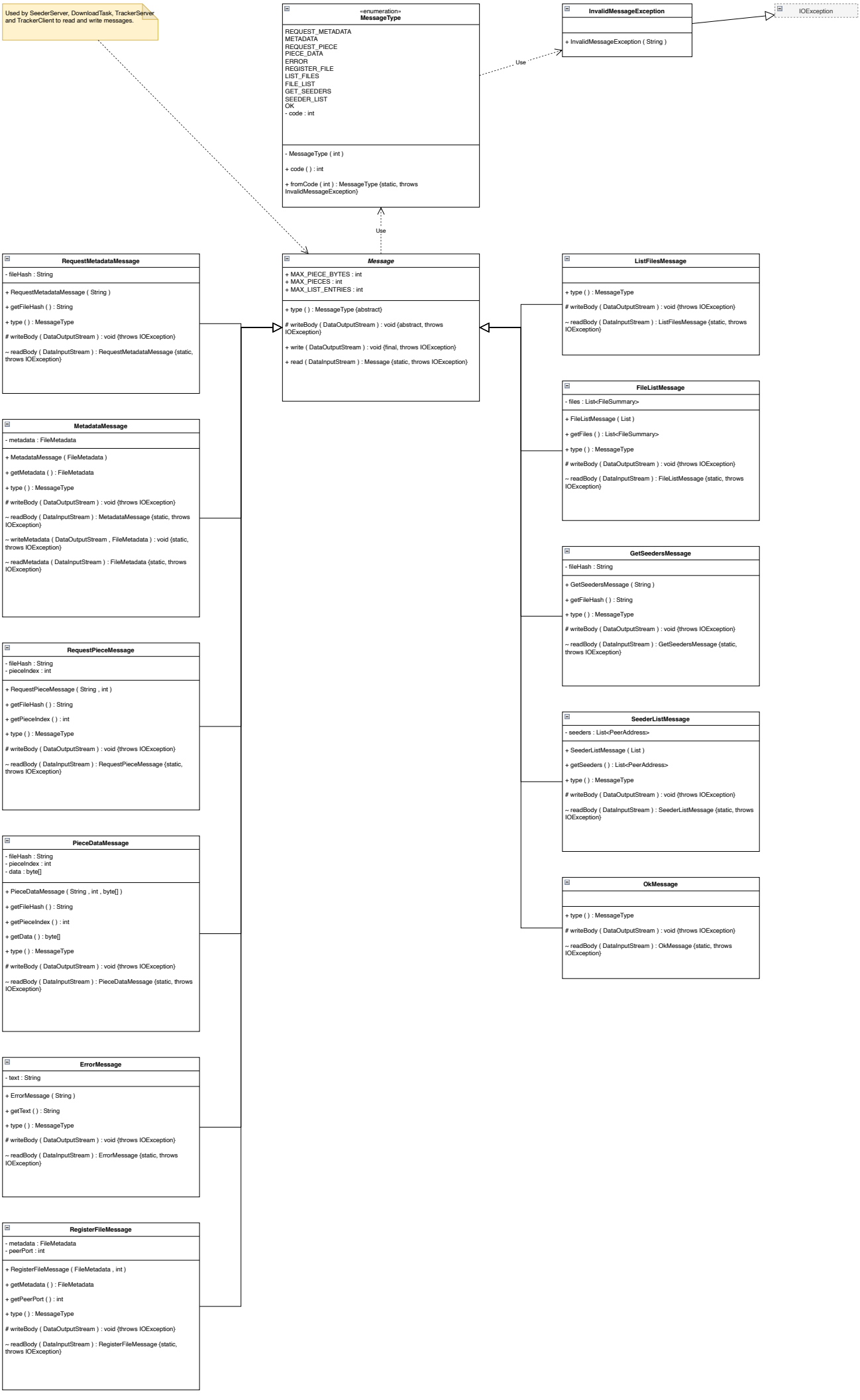
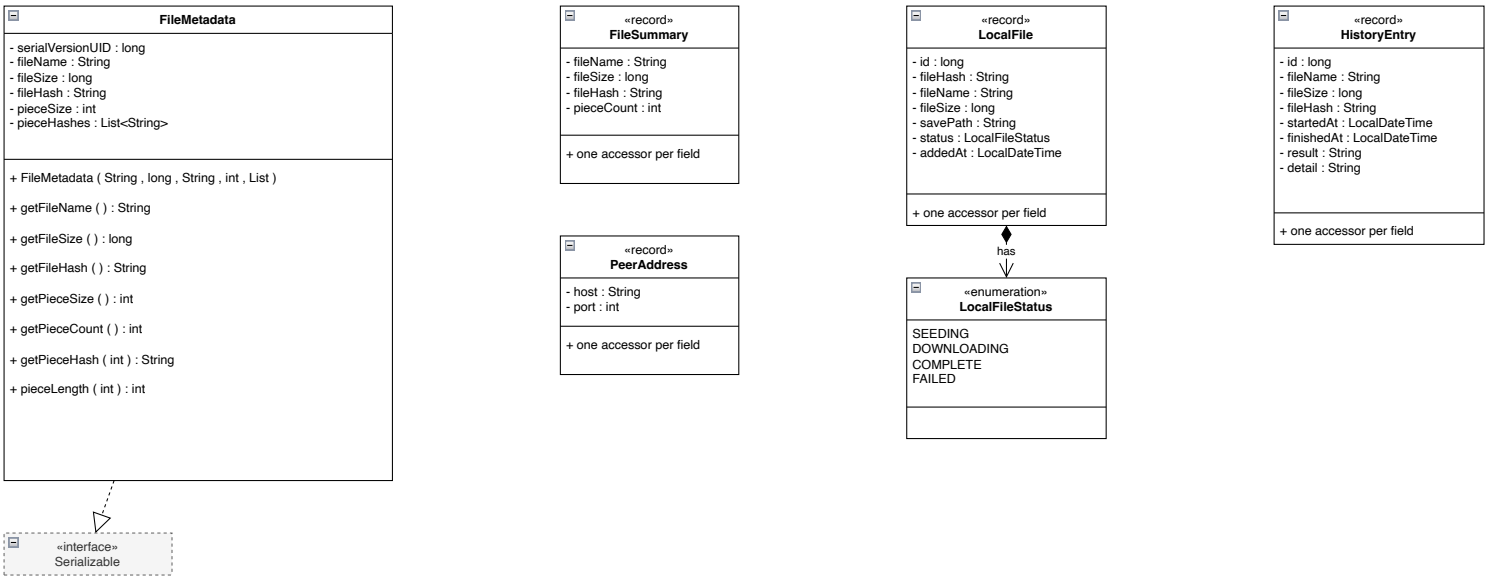

Application (JavaFX)
MainApp
- LOG : Logger
- ctx : AppContext
+ init( ) : void {throws Exception}
+start (Stage) : void
+stop ( ) : void
+ main (String[]) : void
Use
AppContext
- LOG : Logger
+ TRACKER_HOST : String
+ TRACKER_PORT : String
+ PEER_PORT : String
+ DOWNLOAD_DIR : String
- SETTING_KEYS : List<String>
- data : PeerDataService
- settings : Map<String, String>
- dataListeners : List<Runnable>
- executor : ExecutorService
- seeder : SeederServer
+ AppContext ( ) {throws SQLException}
+ data ( ) : PeerDataService
+ setting (String) : String
+ intSetting (String) : int
+ saveSettings (Map) : void {throws SQLException}
+ tracker ( ) : TrackerClient
+ trackerAddress ( ) : String
+ downloadDir ( ) : Path
- seeder ( ) : SeederServer {throws IOException}
+ seedFile (Path) : SeedResult {throws IOException, SQLException}
+ resumeSeeding ( ) : void
+ execute (Runnable) : void
+ runAsync (Callable , Consumer , Consumer) : void
+ addDataListener (Runnable) : void
+ fireDataChanged ( ) : void
+ shutdown ( ) : void
TrackerClient
SeederServer
PeerDataService
Use
«record»
SeedResult
- meta : FileMetadata
- trackerWarning : String
+ one accessor per field
Tables
- Tables ( )
~ textColumn (String , double , Function) : TableColumn {static}
Dialogs
- Dialogs ( )
~ info (String , String) : void {static}
~ warning (String , String) : void {static}
~ error (String , String) : void {static}
~ confirm (String , String) : boolean {static}
~ messageOf (Throwable) : String {static}
- show (AlertType , String , String) : void {static}
AvailableFilesTab
- ctx : AppContext
- downloads : DownloadsTab
- table : TableView<FileSummary>
- status : Label
+ AvailableFilesTab (AppContext , DownloadsTab)
+ refresh (boolean) : void
- downloadSelected ( ) : void
DownloadsTab
- ctx : AppContext
- items : ObservableList<DownloadItem>
+ DownloadsTab (AppContext)
+ start (FileSummary) : boolean
- finish (DownloadItem , String , double) : void
has
DownloadItem
- fileHash : String
- name : StringProperty
- progress : DoubleProperty
- status : StringProperty
- active : boolean
~ DownloadItem (String , String)
MyFilesTab
- WHEN : DateTimeFormatter
- ctx : AppContext
- table : TableView<LocalFile>
- status : Label
- busy : BooleanProperty
+ MyFilesTab (AppContext)
+ refresh ( ) : void
- chooseAndSeed ( ) : void
- seedSelected ( ) : void
- seed (Path) : void
- removeSelected ( ) : void
HistoryTab
- WHEN : DateTimeFormatter
- ctx : AppContext
- table : TableView<HistoryEntry>
- status : Label
+ HistoryTab (AppContext)
+ refresh ( ) : void
- clear ( ) : void
SettingsTab
- ctx : AppContext
- trackerHost : TextField
- trackerPort : TextField
- downloadDir : TextField
- peerPort : TextField
- status : Label
+ SettingsTab (AppContext)
- chooseFolder ( ) : void
- save ( ) : void
- testConnection ( ) : void
- readValues ( ) : Values
- parsePort (String , String) : int {static}
Tab (JavaFX)

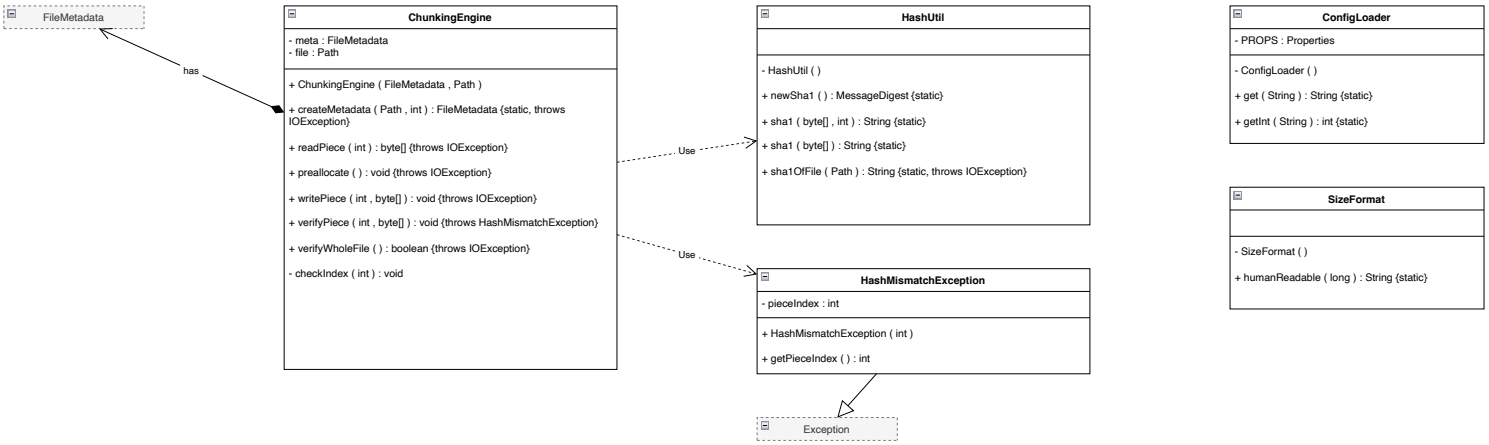

DownloadsTab
TrackerClient
DownloadMain
+ main (String[]) : void {static, throws Exception}
Use
FileDownloader
- LOG : Logger
- FileDownloader ( )
+ downloadFromTracker (TrackerClient , String , Path , ProgressListener) : Path {static, throws IOException}
+ downloadAndRecord (TrackerClient , PeerDataService , String , Path , ProgressListener) : Path {static, throws IOException}
- download (TrackerClient , FileMetadata , Path , ProgressListener) : Path {static, throws IOException}
- record (DbAction) : void {static}
PeerDataService
- db : DatabaseConnection
- localFiles : LocalFileDao
- history : HistoryDao
- settings : SettingsDao
+ PeerDataService (DatabaseConnection)
+ recordSeed (FileMetadata , Path) : void {throws SQLException}
+ beginDownload (FileMetadata , Path) : void {throws SQLException}
+ completeDownload (FileMetadata , Path , LocalDateTime) : void {throws SQLException}
+ failDownload (FileMetadata , Path , LocalDateTime , String) : void {throws SQLException}
- finish (FileMetadata , Path , LocalDateTime , LocalFileStatus , String , String) : void {throws SQLException}
+ listFiles ( ) : List<LocalFile> {throws SQLException}
+ removeFile (long) : boolean {throws SQLException}
+ listHistory ( ) : List<HistoryEntry> {throws SQLException}
+ clearHistory ( ) : int {throws SQLException}
+ getSetting (String , String) : String {throws SQLException}
+ saveSetting (String , String) : void {throws SQLException}
DownloadTask
- LOG : Logger
- MAX_RETRIES : int
- TIMEOUT_MS : int
- host : String
- port : int
- fileHash : String
- trustedMeta : FileMetadata
- downloadDir : Path
- listener : ProgressListener
+ DownloadTask (String , int , String , Path , ProgressListener)
+ DownloadTask (String , int , FileMetadata , Path , ProgressListener)
- DownloadTask (String , int , String , FileMetadata , Path , ProgressListener)
+ download ( ) : Path {throws IOException, HashMismatchException}
+ targetFor (Path , FileMetadata) : Path {static}
- fetchMetadata (DataInputStream , DataOutputStream) : FileMetadata {throws IOException}
- fetchPiece (DataInputStream , DataOutputStream , ChunkingEngine , int) : void {throws IOException, HashMismatchException}
«interface»
ProgressListener
+ onProgress (int , int) : void
ChunkingEngine
SeederServer
- LOG : Logger
- IDLE_TIMEOUT_MS : int
- port : int
- files : Map<String, SharedFile>
- pool : ExecutorService
- serverSocket : ServerSocket
- running : boolean
+ SeederServer (int)
+ share (Path , int) : FileMetadata {throws IOException}
+ start ( ) : void {throws IOException}
+ getPort ( ) : int
+ stop ( ) : void
- acceptLoop ( ) : void
- handle (Socket) : void
- respond (Message) : Message {throws IOException}
«record»
SharedFile
- meta : FileMetadata
- engine : ChunkingEngine
+ one accessor per field
SeederMain
+ main (String[]) : void {static, throws Exception}
has


TrackerMain
+ main (String[]) : void {static, throws Exception}
Use
TrackerServer
- LOG : Logger
- IDLE_TIMEOUT_MS : int
- port : int
- service : TrackerService
- pool : ExecutorService
- serverSocket : ServerSocket
- running : boolean
+ TrackerServer (int , TrackerService)
+ start ( ) : void {throws IOException}
+ getPort ( ) : int
+ stop ( ) : void
- acceptLoop ( ) : void
- handle (Socket) : void
- respond (Message , String) : Message
has
TrackerService
- db : DatabaseConnection
- files : SharedFileDao
- peers : PeerDao
+ TrackerService (DatabaseConnection)
+ registerFile (FileMetadata , String , int) : void {throws SQLException}
+ listFiles ( ) : List<FileSummary> {throws SQLException}
+ getMetadata (String) : Optional<FileMetadata> {throws SQLException}
+ getSeeders (String) : List<PeerAddress> {throws SQLException}
has
DatabaseConnection
has
SharedFileDao
has
PeerDao
TrackerClient
- TIMEOUT_MS : int
- host : String
- port : int
+ TrackerClient (String , int)
+ registerFile (FileMetadata , int) : void {throws IOException}
+ listFiles ( ) : List<FileSummary> {throws IOException}
+ getMetadata (String) : FileMetadata {throws IOException}
+ getSeeders (String) : List<PeerAddress> {throws IOException}
- call (Message , Class<T>) : T {throws IOException}
Use
Use
Message


DatabaseConnection
- TRACKER_SCHEMA : String[]
- PEER_SCHEMA : String[]
- url : String
- schema : String[]
+ DatabaseConnection (String)
- DatabaseConnection (String , String[])
+ forPeer (String) : DatabaseConnection {static}
+ open ( ) : Connection {throws SQLException}
+ initSchema ( ) : void {throws SQLException}
SharedFileDao
+ insertFile (Connection , FileMetadata) : boolean {throws SQLException}
+ findIdByHash (Connection , String) : Optional<Long> {throws SQLException}
+ insertPieces (Connection , long , FileMetadata) : void {throws SQLException}
+ listAll (Connection) : List<FileSummary> {throws SQLException}
+ findMetadata (Connection , String) : Optional<FileMetadata> {throws SQLException}
TrackerService
PeerDao
+ upsertPeer (Connection , String , int) : long {throws SQLException}
+ addSeed (Connection , long , long) : void {throws SQLException}
+ findSeeders (Connection , String) : List<PeerAddress> {throws SQLException}
LocalFileDao
- normalize (Path) : String {static}
+ upsert (Connection , FileMetadata , Path , LocalFileStatus) : void {throws SQLException}
+ updateStatus (Connection , String , Path , LocalFileStatus) : int {throws SQLException}
+ findAll (Connection) : List<LocalFile> {throws SQLException}
+ delete (Connection , long) : boolean {throws SQLException}
PeerDataService
HistoryDao
+ insert (Connection , FileMetadata , LocalDateTime , LocalDateTime , String , String) : void {throws SQLException}
+ findAll (Connection) : List<HistoryEntry> {throws SQLException}
+ deleteAll (Connection) : int {throws SQLException}
SettingsDao
+ get (Connection , String) : Optional<String> {throws SQLException}
+ set (Connection , String , String) : void {throws SQLException}
has
has
has
has
has
has
has
has


Used by SeederServer, DownloadTask, TrackerServer and TrackerClient to read and write messages.
«enumeration»
MessageType
REQUEST_METADATA
METADATA
REQUEST_PIECE
PIECE_DATA
ERROR
REGISTER_FILE
LIST_FILES
FILE_LIST
GET_SEEDERS
SEEDER_LIST
OK
- code : int
- MessageType (int)
+ code ( ) : int
+ fromCode (int) : MessageType {static, throws InvalidMessageException}
Use
InvalidMessageException
+ InvalidMessageException (String)
IOException
Use
Message
+ MAX_PIECE_BYTES : int
+ MAX_PIECES : int
+ MAX_LIST_ENTRIES : int
+ type ( ) : MessageType {abstract}
writeBody (DataOutputStream) : void {abstract, throws IOException}
+ write (DataOutputStream) : void {final, throws IOException}
+ read (DataInputStream) : Message {static, throws IOException}
RequestMetadataMessage
- fileHash : String
+ RequestMetadataMessage (String)
+ getFileHash ( ) : String
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : RequestMetadataMessage {static, throws IOException}
MetadataMessage
- metadata : FileMetadata
+ MetadataMessage (FileMetadata)
+ getMetadata ( ) : FileMetadata
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : MetadataMessage {static, throws IOException}
~ writeMetadata (DataOutputStream , FileMetadata) : void {static, throws IOException}
~ readMetadata (DataInputStream) : FileMetadata {static, throws IOException}
RequestPieceMessage
- fileHash : String
- pieceIndex : int
+ RequestPieceMessage (String , int)
+ getFileHash ( ) : String
+ getPieceIndex ( ) : int
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : RequestPieceMessage {static, throws IOException}
PieceDataMessage
- fileHash : String
- pieceIndex : int
- data : byte[]
+ PieceDataMessage (String , int , byte[])
+ getFileHash ( ) : String
+ getPieceIndex ( ) : int
+ getData ( ) : byte[]
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : PieceDataMessage {static, throws IOException}
ErrorMessage
- text : String
+ ErrorMessage (String)
+ getText ( ) : String
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : ErrorMessage {static, throws IOException}
RegisterFileMessage
- metadata : FileMetadata
- peerPort : int
+ RegisterFileMessage (FileMetadata , int)
+ getMetadata ( ) : FileMetadata
+ getPeerPort ( ) : int
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : RegisterFileMessage {static, throws IOException}
ListFilesMessage
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : ListFilesMessage {static, throws IOException}
FileListMessage
- files : List<FileSummary>
+ FileListMessage (List)
+ getFiles ( ) : List<FileSummary>
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : FileListMessage {static, throws IOException}
GetSeedersMessage
- fileHash : String
+ GetSeedersMessage (String)
+ getFileHash ( ) : String
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : GetSeedersMessage {static, throws IOException}
SeederListMessage
- seeders : List<PeerAddress>
+ SeederListMessage (List)
+ getSeeders ( ) : List<PeerAddress>
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : SeederListMessage {static, throws IOException}
OkMessage
+ type ( ) : MessageType
writeBody (DataOutputStream) : void {throws IOException}
~ readBody (DataInputStream) : OkMessage {static, throws IOException}


FileMetadata
- serialVersionUID : long
- fileName : String
- fileSize : long
- fileHash : String
- pieceSize : int
- pieceHashes : List<String>
+ FileMetadata (String , long , String , int , List)
+ getFileName ( ) : String
+ getFileSize ( ) : long
+ getFileHash ( ) : String
+ getPieceSize ( ) : int
+ getPieceCount ( ) : int
+ getPieceHash (int) : String
+ pieceLength (int) : int
«interface»
Serializable
«record»
FileSummary
- fileName : String
- fileSize : long
- fileHash : String
- pieceCount : int
+ one accessor per field
«record»
PeerAddress
- host : String
- port : int
+ one accessor per field
«record»
LocalFile
- id : long
- fileHash : String
- fileName : String
- fileSize : long
- savePath : String
- status : LocalFileStatus
- addedAt : LocalDateTime
+ one accessor per field
has
«enumeration»
LocalFileStatus
SEEDING
DOWNLOADING
COMPLETE
FAILED
«record»
HistoryEntry
- id : long
- fileName : String
- fileSize : long
- fileHash : String
- startedAt : LocalDateTime
- finishedAt : LocalDateTime
- result : String
- detail : String
+ one accessor per field


FileMetadata
has
ChunkingEngine
- meta : FileMetadata
- file : Path
+ ChunkingEngine (FileMetadata , Path)
+ createMetadata (Path , int) : FileMetadata {static, throws IOException}
+ readPiece (int) : byte[] {throws IOException}
+ preallocate ( ) : void {throws IOException}
+ writePiece (int , byte[]) : void {throws IOException}
+ verifyPiece (int , byte[]) : void {throws HashMismatchException}
+ verifyWholeFile ( ) : boolean {throws IOException}
- checkIndex (int) : void
Use
Use
HashUtil
- HashUtil ( )
+ newSha1 ( ) : MessageDigest {static}
+ sha1 (byte[] , int) : String {static}
+ sha1 (byte[]) : String {static}
+ sha1OfFile (Path) : String {static, throws IOException}
HashMismatchException
- pieceIndex : int
+ HashMismatchException (int)
+ getPieceIndex ( ) : int
Exception
ConfigLoader
- PROPS : Properties
- ConfigLoader ( )
+ get (String) : String {static}
+ getInt (String) : int {static}
SizeFormat
- SizeFormat ( )
+ humanReadable (long) : String {static}